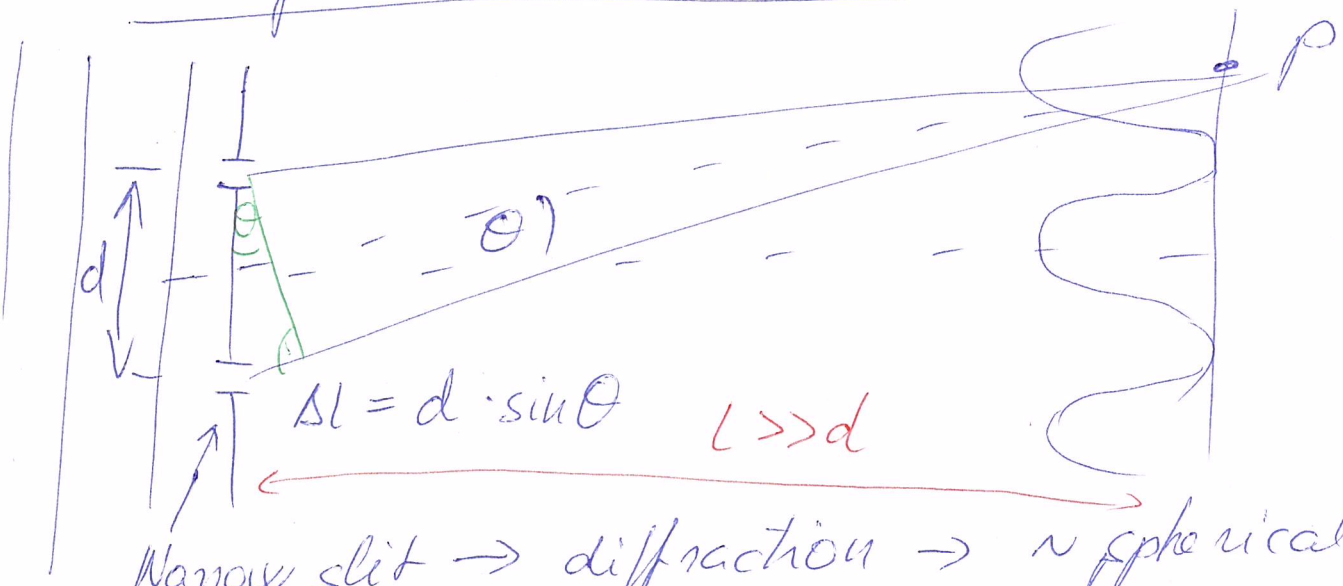
Interference & Diffraction (18)

Young's double slit exp.



Narrow slit → diffraction → ~ spherical wave

$$E = E_1 + E_2 = E_0 \cdot \frac{e^{-i\omega t}}{r}\left(e^{ik\left(r-\frac{\Delta l}{2}\right)} + e^{ik\left(r+\frac{\Delta l}{2}\right)}\right)$$

$$= E_0 \cdot \frac{e^{i(kr-\omega t)}}{r} \cdot 2\cos\left(\frac{k\Delta l}{2}\right)$$

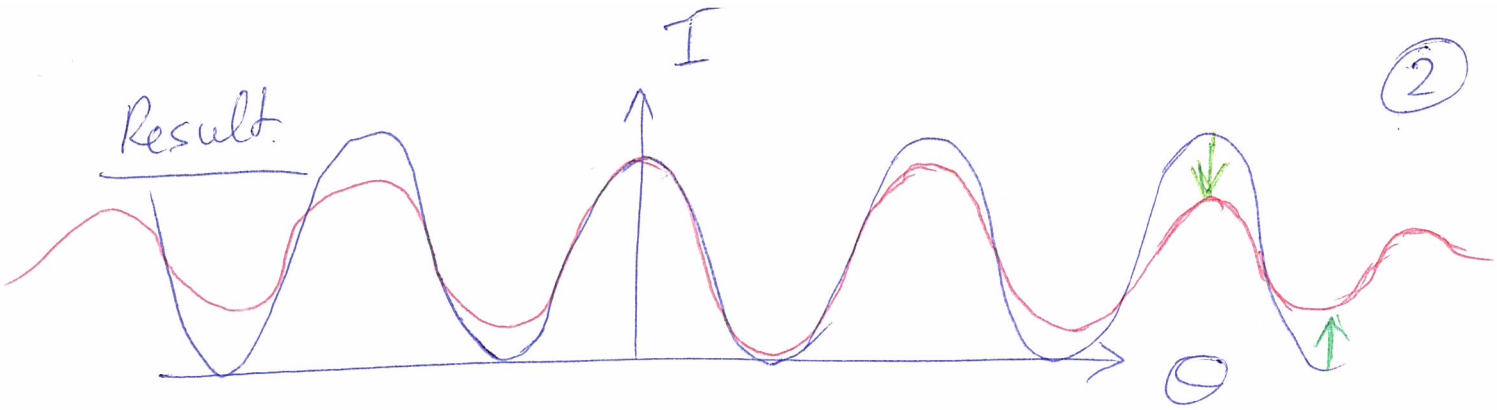
$$\langle |S| \rangle = I = c\varepsilon\langle E^2\rangle = c\varepsilon \frac{4E_0^2}{r^2} \cdot \cos^2\left(\frac{k\Delta l}{2}\right)$$

$$= \text{const} \cdot \cos^2\left(\frac{k d \cdot \sin\theta}{2}\right)$$

Places of constructive interference (maxima):

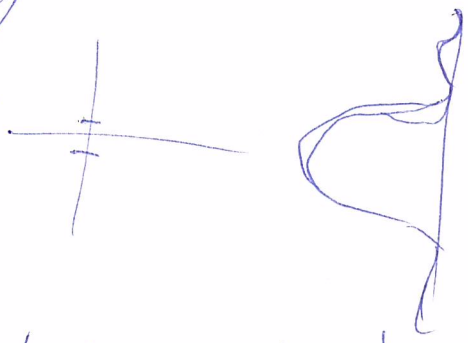
$$\frac{k d \sin\theta}{2} = n\pi \rightarrow \frac{2\pi}{\lambda \cdot 2} \cdot d \sin\theta = n\pi$$

$$\Delta l = \boxed{d \sin\theta = n\lambda}$$



- Maxima reduced for larger angles:

finite aperture size → diffraction

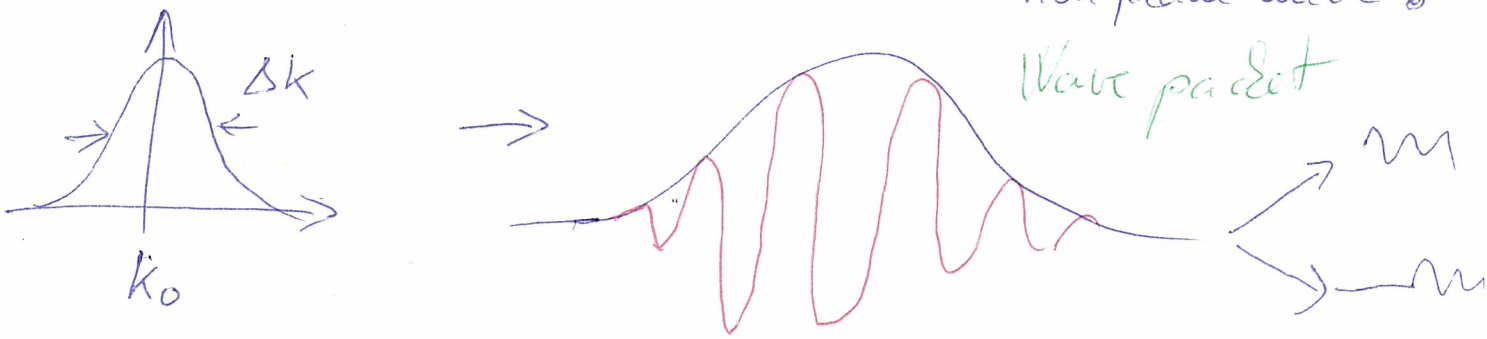


- Minima growing: (contrast decreasing)

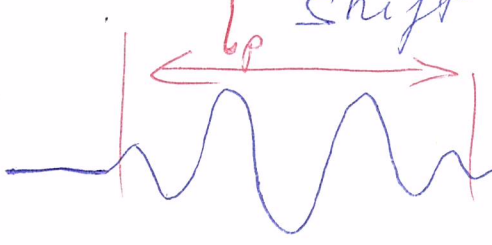
↓

Finite coherence length of light source

Coherence (temporal) of a light source: real light: non plane wave!



Double slit: split wave into 2, shift one packet to the other



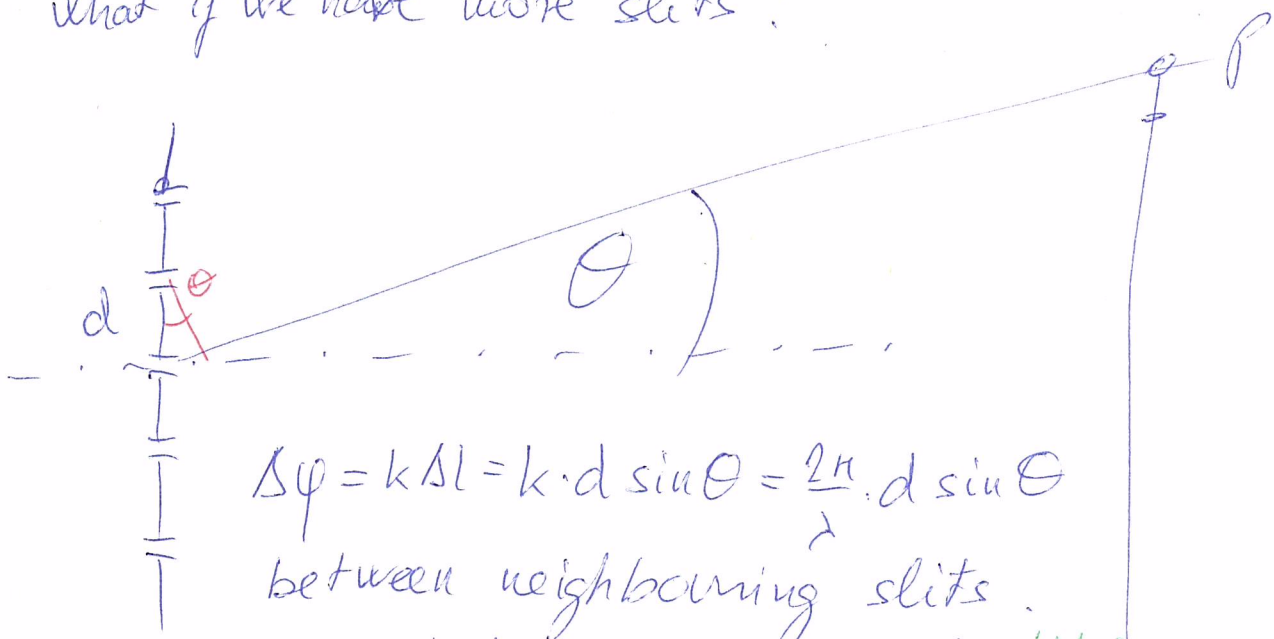
$\Delta l = 1 \cdot \lambda$ → constructive

no interference!



$\Delta L = 10\lambda > l_p = l_{coh}$

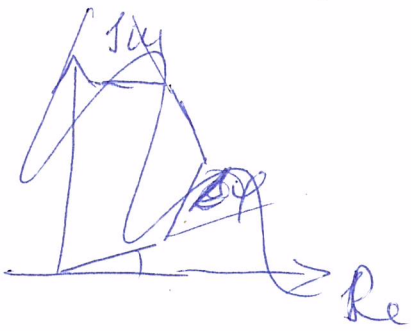
What if we have more slits?

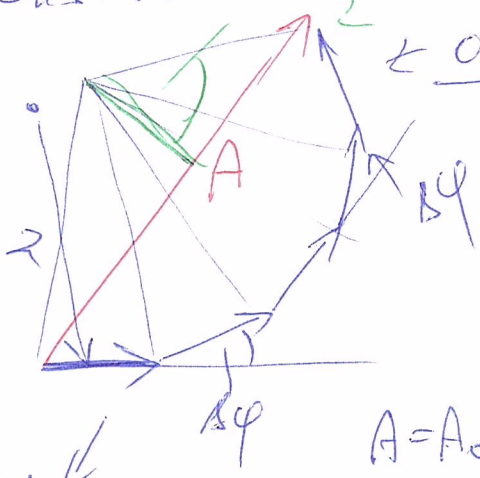


$\Delta\varphi = k\Delta l = k \cdot d \sin\theta = \frac{2\pi}{\lambda} \cdot d \sin\theta$

between neighbouring slits.

Amplitudes are constant: $\frac{N\Delta\varphi}{2}$





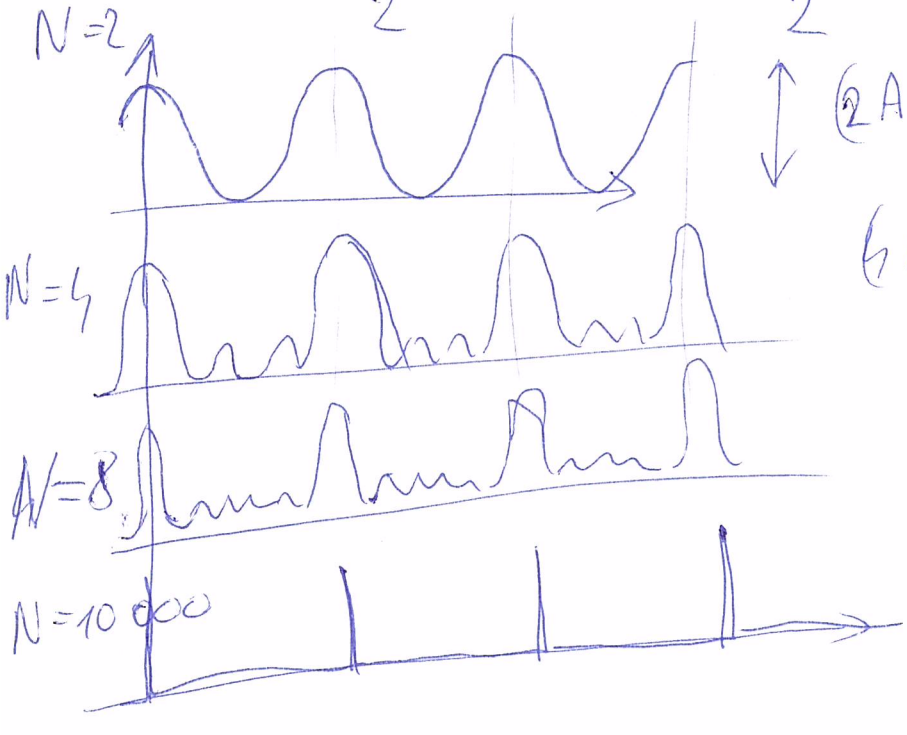
← on a circle

$A = 2r \sin\frac{N\Delta\varphi}{2}$ →

$A = A_0 \frac{\sin\left(\frac{N\Delta\varphi}{2}\right)}{\sin\frac{\Delta\varphi}{2}}$

$\frac{A_0}{2} = r \cdot \sin\frac{\Delta\varphi}{2} \Rightarrow r = \frac{A_0}{2}\frac{1}{\sin\frac{\Delta\varphi}{2}}$

$\left(\frac{\sin 2x}{\sin x} = \frac{2\sin\cdot\cos}{\sin}\right)$

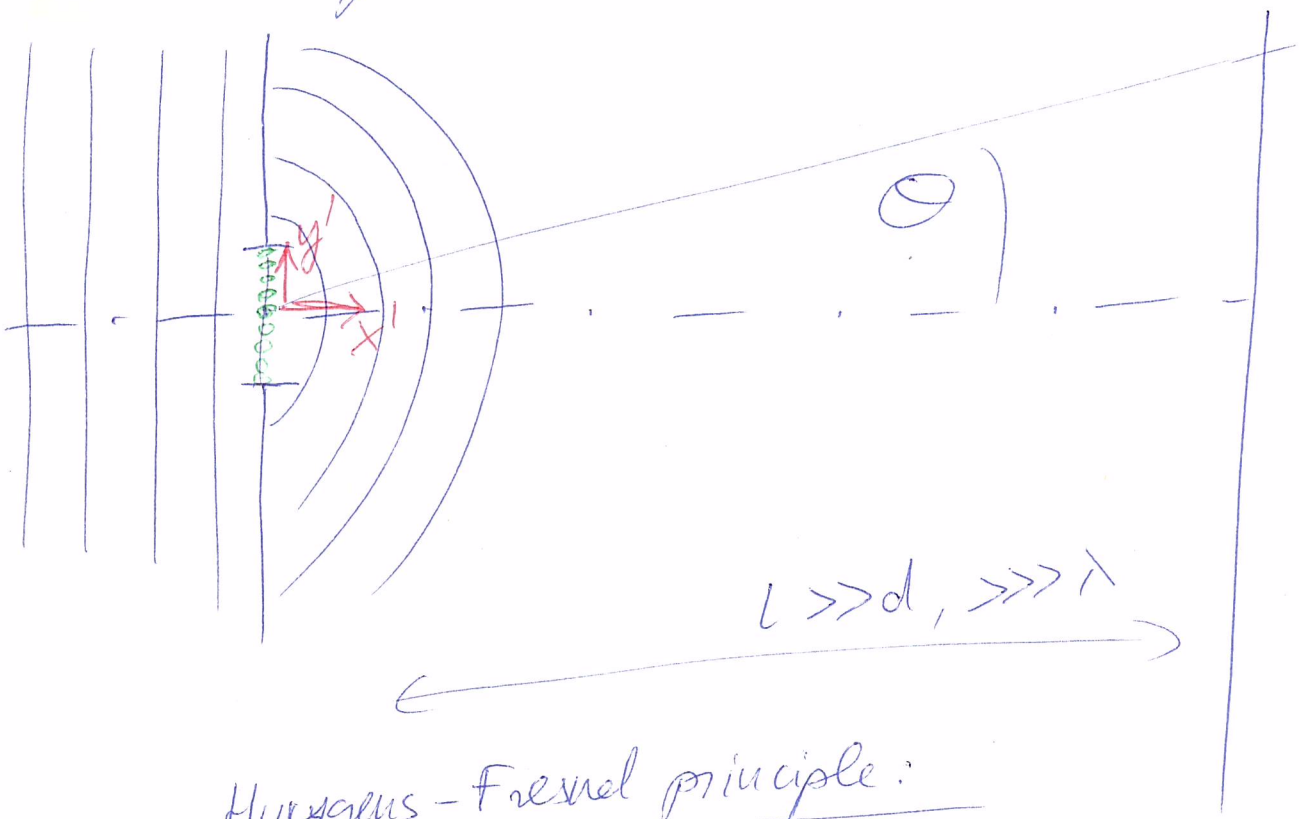


Diffraction grating

$d \sin\theta = n\lambda$

← Fourier transform of ⊓⊓⊓⊓⊓

Let's investigate the diffraction on a single slit!



Huygens-Fresnel principle:

The new wavefront is the sum of elementary sph. "wavelets". (More rigorously we have to solve the boundary value problem...)

A single sph. wave:

$$E(y',\Theta) = \frac{E_0}{r} \cdot e^{-i\omega t} \cdot e^{ik\cos\Theta x} \; e^{ik\sin\Theta (y-y')}$$

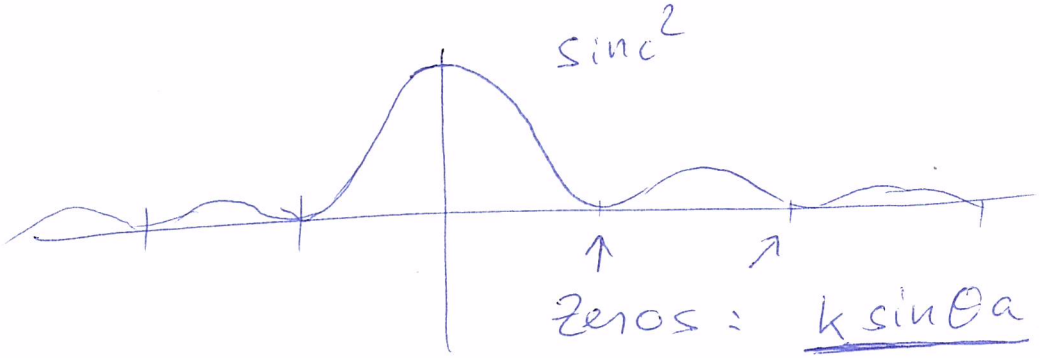
Summing up for $-\frac{a}{2} \to \frac{a}{2}$

$$E(\Theta) = \int_{-\frac{a}{2}}^{\frac{a}{2}} \frac{E_0}{r} \cdot e^{i(k\cos\Theta x + k\sin\Theta y - \omega t)} \cdot e^{-ik\sin\Theta y'} \, dy'$$

$$= \frac{E_0}{r} \cdot e^{i(kr-\omega t)} \cdot \left[\frac{e^{-ik\sin\Theta\frac{a}{2}} - e^{ik\sin\Theta\frac{a}{2}}}{-ik\sin\Theta} \right] =$$

$$= \frac{E_0}{r} e^{i(kr-\omega t)} \cdot 2 \sin \frac{k \sin\theta a}{2} \Big/ k \sin\theta a \quad \leftarrow \quad \frac{\sin x}{x} = \operatorname{sinc} x$$

$$I(\theta) = \frac{E_0^2}{r^2} \cdot C \cdot \operatorname{sinc}^2\left(\frac{k \sin\theta a}{2}\right)$$

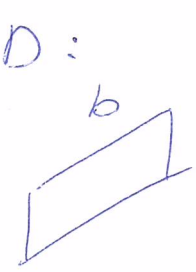


$$\text{zeros}: \quad \frac{k \sin\theta a}{2} = n\pi \quad (\text{except } n=0)$$

$$\sin\theta = \frac{n\,2\pi}{k\,a} = n \cdot \frac{\lambda}{a}$$

The smaller the slit is the larger the diff. angle!

In 2D:



$$\rightarrow E(\theta,\varphi) = \int_{-\frac{a}{2}}^{\frac{a}{2}} \left(\int_{-\frac{b}{2}}^{\frac{b}{2}} E_0 e^{-iky' \sin\theta} dy' \right) \cdot e^{-ikz' \sin\varphi} dz'$$

1D Fourier transform!

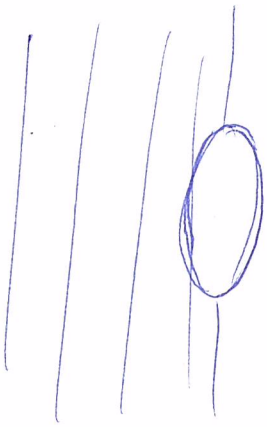
2D FT ← generally truth.

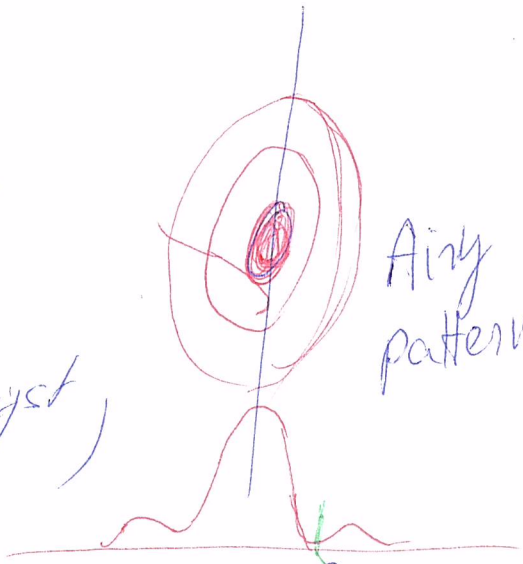
$$\operatorname{sinc}^2\left(\frac{k \sin\theta a}{2}\right) \cdot \operatorname{sinc}^2\left(\frac{k \sin\varphi b}{2}\right)$$

Fourier optics!



For a circular aperture:





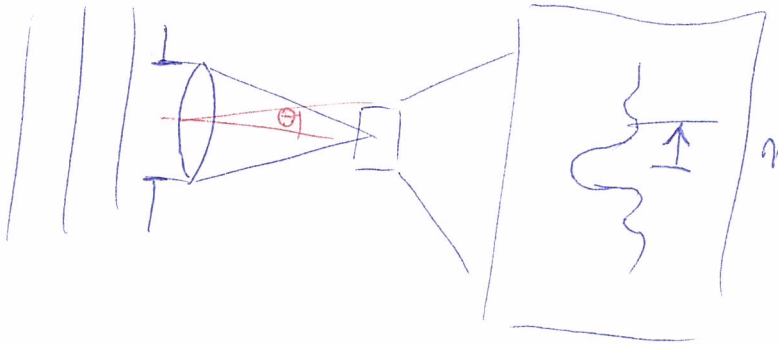
Integration in cyl. Coordsyst
(or 2DFT)

$$I(\Theta) = E_0^2 \cdot \left[2 \left| \frac{J_1\left(k \frac{d}{2} \sin\Theta\right)}{k \frac{d}{2} \sin\Theta} \right| \right]^2$$

$J_1(x)$: Bessel function of the 1st kind.

First $\varnothing$ $\sin\Theta = 1.22 \frac{\lambda}{d}$

The smaller d the larger the spot size!



$r_0 = 1{,}22 \frac{\lambda}{d} \cdot f$ diffraction limit of spot size

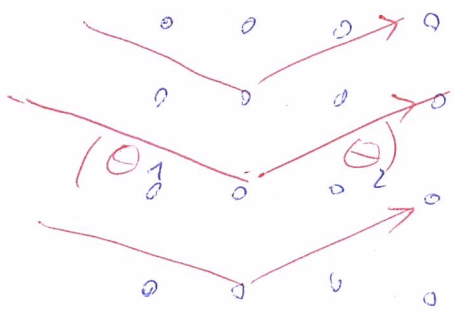
Rayleigh resolution criterium:

if a dual-star has an angular separation larger than ~~the~~ r_0 ~~focal~~ $\sim \frac{\lambda}{d}$)

Even more important: at const f, the luminosity of a star is $\sim d^4$ (large telescopes!!!) see further

Resolution of a camera, lithographic system everything is behaving like this!

Bragg diffraction:



Constructive interference for multiple layers!

← like reflection on planes

$\theta_1 = \theta_2$

$$m\lambda = 2d \sin\theta$$

Investigation of cristals

3D index modulation, hologram!